Projection by Convolution

Optimal Sample Complexity for Reinforcement Learning in Continuous-Space MDPs

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Mismatch between theory and practice

Tabular MDPs

$$\mathcal{S},\ \mathcal{A} \text{ finite}$$

Linear MDPs

$$p_h(s'|s,a) = \boldsymbol{\varphi}(s,a)^\top \boldsymbol{\mu}_h(s')$$

$$R_h(s,a) = \boldsymbol{\varphi}(s,a)^\top \boldsymbol{\theta}_h$$

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$$\mathcal{S} = [-1,1]^{d_S}, \mathcal{A} = [-1,1]^{d_A}$$

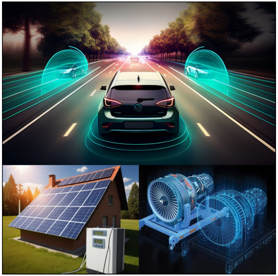

Figure: Practical RL with continuous state-action space

Smoothness

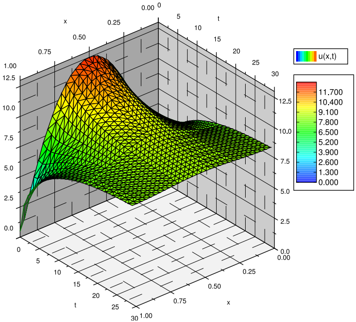

Figure: Solution of heat equation

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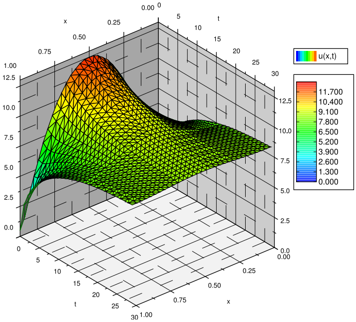

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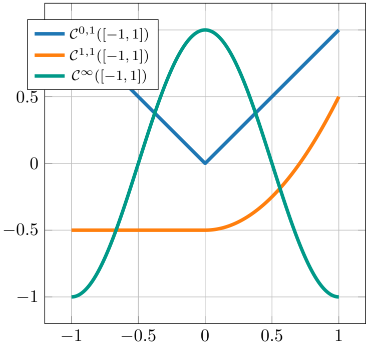

Figure: Examples of functions $\mathcal{C}^{\nu,1}(\mathcal{S} \times \mathcal{A})$: ν continuous derivative, and the last one is Lipschitz continuous

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Bellmann optimality operator

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$$\mathcal{T}_h f(s,a) := R_h(s,a) + \mathbb{E}_{s' \sim p_h(\cdot|s,a)}\Big[\sup_{a' \in \mathcal{A}} f(s',a')\Big].$$

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$\nu-$**Smooth MDP:** $\mathcal{T}_h$ is bounded on $\mathcal{C}^{\nu,1}(\mathcal{S} \times \mathcal{A}) \to \mathcal{C}^{\nu,1}(\mathcal{S} \times \mathcal{A})$

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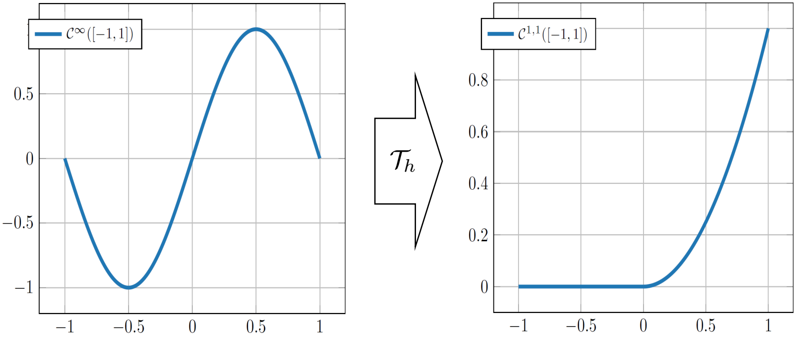
$\|f\|_{C^{\nu,1}}$ supremum of the first $\nu+1$ derivatives.
Meaning: **Operator maintains smoothness up to derivative** $\nu-$**th.**

Example

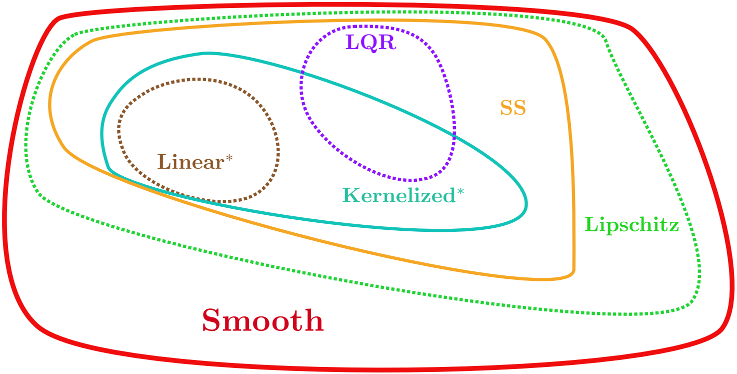
Consider the case $d = 1$



For which values of ν this process can be smooth? $\nu = 1$, not $\nu = 2$.

All studied settings for continuous MDPs are simpler [Maran et al., 2024]

1. Kernelized assumes p_h, R_h belong to Matérn Kernel RKHS
2. Strongly Smooth assumes p_h, R_h to be smooth functions



Periodicity

$p_h(\cdot|s,a), R_h(s,a)$ achieve the same value at the boundaries of $\mathcal{S} \times \mathcal{A}$

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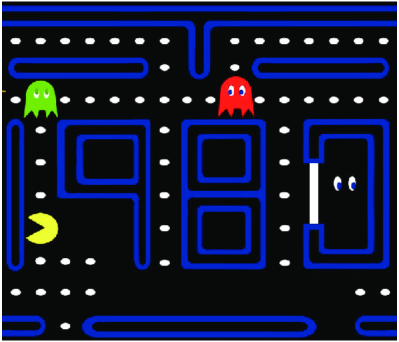

Power of trigonometric approximation

Use a feature map $\varphi_N : \mathcal{S} \times \mathcal{A} \to \mathbb{R}^{\widetilde{N}}$ of trigonometric components.

$$\varphi_N(s,a) = [1, \sin(s), \cos(s) \dots] \qquad \widetilde{N} = \binom{2N+d}{d} \leq (2N+1)^d.$$

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Theorem [Schultz, 1969]

For every $f \in \mathcal{C}^{\nu,1}([-1,1]^d)$ exists parameter $\boldsymbol{\theta} \in \mathbb{R}^{\widetilde{N}}$ s.t.

$$\|f(\cdot) - \varphi_N(\cdot)^\top \boldsymbol{\theta}\|_{L^\infty} \leq C\|f\|_{\mathcal{C}^{\nu,1}} \mathbf{N}^{-(\nu+1)}$$

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Approximate Q_h^* function

$$Q_h^*(s,a) = \varphi_N(s,a)^\top \boldsymbol{\theta}_h + \xi_N(s,a).$$

Standard solution: LSVI

LSVI:

1. Start with $\widehat{V}_{H+1}(s) = 0$
2. Choose $\{(s_i, a_i)\}_{i=1,\dots,n}$ from **optimal design** and simulate $s_i' \sim p_h(\cdot|s_i, a_i)$, $r_i \sim R_h(s_i, a_i)$
3. Fit $\boldsymbol{\theta}_h$ to minimize

$$\sum_{i=1}^{n} (\varphi_N(s_i, a_i)^\top \boldsymbol{\theta}_h - r_i - \widehat{V}_{h+1}(s_i'))^2$$

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Not so fast!

Existing algorithms [Lattimore et al., 2020, Zanette et al., 2020], only guarantee

$$\|\widehat{Q}_h(\cdot,\cdot) - Q_h^*(\cdot,\cdot)\|_{L^\infty} \leq \underbrace{\sqrt{\widetilde{N}}}_{\textbf{amplification}} \underbrace{\|\xi_N(\cdot)\|_{L^\infty}}_{miss.} + o(1)$$

New regression method

1. Guarantee on $\|f - \widehat{f}_n\|_{L^\infty}$
2. Immune to misspecification
3. Approximate the derivatives of the function

Avoid misspecification $\implies$ **Project** target function

$$\mathcal{P}_{N,\infty}[f] = \underset{\boldsymbol{\theta} \in \mathbb{R}^{\widetilde{N}}}{\arg\min} \|f(\cdot) - \varphi_N(\cdot)^\top \boldsymbol{\theta}\|_{L^\infty}$$

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$$\boxed{\text{But we don't know } f(\cdot)!}$$

$$x_n \to y_n = f(x_n) + \zeta_n.$$

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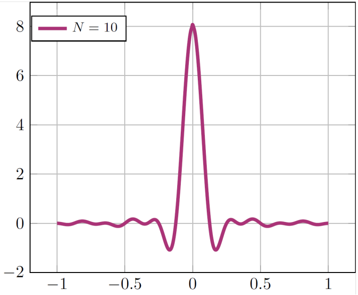

Figure: De la Vallée Poussin kernel $D_N(\cdot)$

Theorem

Let $f \in \mathcal{C}_p^{\nu,1}([-1,1]^d)$. Then, $D_N * f \in \mathbb{T}_N$ is a trigonometric polynomial, and we have

$$\|f - D_N * f\|_{L^\infty} \leq \frac{2^{\nu+2}\Lambda_d C}{\mathbf{N}^{\nu+1}} \|f\|_{\mathcal{C}^{\nu,1}}$$

Convolution $=$ Noise

Learner asks for $x_i \in [-1,1]^d$, but we query $x_i' = x_i + \eta$, with $\eta \sim D_N(\cdot)$

$$\mathbb{E}[y_i] = \mathbb{E}[f(x_i + \eta)] = D_N * f(x_i).$$

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Regression bound

As the target is purely linear, we only pay the approximation error

$$\|\widehat{f}_n(\cdot) - f(\cdot)\|_{L^\infty} \leq C\|\xi_N(\cdot)\|_{L^\infty} + \widetilde{\mathcal{O}}((\widetilde{N}/n)^{1/2}).$$

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What to do with LSVI?

1. Sample η_+,η_- from pos./neg. part of $D_N(\cdot)$
2. Query for $s_{h+1}^+ \sim p_h(\cdot|z_h + \eta_+)$
3. Query for $s_{h+1}^- \sim p_h(\cdot|z_h + \eta_-)$
4. $V_{h+1}^{\text{target}}(s_h, a_h) = \big(\beta_+ \widehat{V}_{h+1}(s_{h+1}^+) - \beta_- \widehat{V}_{h+1}(s_{h+1}^-)\big)$

Result

Main theorem

Setting $N = \mathcal{O}(\varepsilon^{-\frac{1}{\nu+1}})$, we learn a $\varepsilon-$optimal policy with

$$n = \widetilde{\mathcal{O}}\left(K(H,d)\varepsilon^{-\frac{d+2(\nu+1)}{\nu+1}}\right)$$

samples w.h.p. with $K(H,d) \approx \exp(dH)$.

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Lower bounds:

- Optimal order in ε for noisy optimization [Wang et al., 2018].
- Exponential dependence in dH for Lipschitz MDPs [Maran et al., 2024].

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Are upper and lower bounds comparable?

We can re-scale hard instances to get periodicity.